

Case Study 1 - Atari

Student submission

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Worksheet link

<https://courses.ethereallab.app/assignment/v4/IT265-001-F2026/case-study-1-atari/dbc>

Grading link

<https://courses.ethereallab.app/assignment/v4/IT265-001-F2026/case-study-1-atari/grading/dbc>

AI disclosure: Completed; 1 entry; 1 declared AI use; 0 declared no AI use; policy v1

Game Overview

Game details

Response prompt

Identify the game: title, developer, publisher, original platform, genre, and release date. Note the version you played if different.

Submitted response

Title: Jr. Pac-Man

Developer: General Computer Corporation

Publisher: Bally Midway

Genre: Maze

Release Date: August 13, 1983

https://www.retrogames.cz/play_190-Atari2600.php

History and industry

Response prompt

Explain the historical context of the release and where the game fit in the industry at that time.

Submitted response

Jr. Pac-Man, similarly to Ms. Pac-Man, was originally created due to the massive success of the original Pac-Man game. However, its release coincided with the year North American video game crash of 1983. Too many competing consoles were sold at stores, and the game itself was facing tensions with Namco, as it was developed without prior permission. Actual Jr. Pac-Man game cabinets were very scarce, and often actually repurposed "Mappy" game cabinets with stickers overlayed on them.

https://youtu.be/DI7w_1Z4a7g?si=LRd7CbeaKTWpyj5_

<https://www.arcade-museum.com/Videogame/jr-pac-man>

Contemporary comparison

Response prompt

Compare the game with a similar game from the same period, explaining a meaningful design similarity or difference.

Submitted response

While technically released prior in the previous year, "Dig Dug" had its Arcade/Atari ports release in 1983. The two games feature a similar play style: a top-down view as the player navigates a grid in order to complete the level. Also similarly, the two deal with monsters, with both having methods to defeat them in order to make the game easier for the player.

Their methods in doing so however are quite different. Like its predecessors, Jr. Pac-Man can only defeat the enemies

(ghosts) once he eats a power pellet, allowing him to eat the ghosts. Dig Dug instead has the ability to inflate them until they burst or crush them underneath rocks. In the latter method both are similar in that they are situational, but Jr. Pac-Man lacks a secondary method to defeating the ghosts.

Dig Dug's enemies are also different. Dig Dug's enemies have the ability to move through walls in order to reach his current tunnel. Ironically, the ghosts in Jr. Pac-Man (who are by nature intangible) cannot.

Lastly, the method in which both move. They both have the ability to move any 4 directions: up, down, left, and right. However, Jr. Pac-Man is restricted to the tunnels that already exist. Dig Dug has the ability to create his own tunnels, and as such has less restriction on his range of movement. The only thing restricting his movement in the game itself are the borders of the screen and the boulders, which he cannot dig through.

Gameplay Mechanics And Level Design

Core gameplay mechanics

Response prompt

Explain how the core actions and rules shape a recurring player decision. Support the explanation with a gameplay example.

Submitted response

Just as its predecessors, Jr. Pac-Man's core actions are very limited. Jr. Pac automatically moves forward in any direction, and the player inputs direct him either up, down, left, or right. In addition to this, the ghosts chasing you as you move are also always on the move, wandering the maze in attempts to catch and defeat you before you can complete your objective. The game is constantly in motion, and incentivizes you to adapt quickly to the movements of the ghosts in order to win. That said, the player also has a way to "fight back" against the ghosts, so to speak. Power Pellets grant Jr. Pac the ability to eat the ghosts in return, with the ghosts later on returning once revisiting the center of the maze. These Power Pellets however are limited to only 6, dropping down to 4 in the later levels of the game.

These features makes the player think about what to do in a given situation. For example, the player could be collecting pellets in one corner of the maze. They are suddenly approached by two ghosts. There is a route in the maze to get away from them, but there is also a Power Pellet nearby. Eating ghosts grants more points, totaling to a higher score overall. Will the player risk not getting to the Power Pellet in time for that higher score, or will they play it safe and move on to a different area? If they choose the latter, then not only do they escape, but they also save the Power Pellet for a more desperate situation.

https://youtu.be/DI7w_1Z4a7g?si=hXMbp4SEqrm4qbob
https://youtu.be/g9n2Zbs_XLI?si=f86h8B8kTvDRD6oT

Level design

Response prompt

Explain how the layout, obstacles, or progression affect a player decision or challenge. Use a specific example.

Submitted response

Layout, obstacles, and progression are almost identical to its original inspiration, Pac-Man. The goal in all given levels is to collect all Pellets throughout the maze. The maze itself is quite vast and constricting. If the player isn't careful, they may find themselves blocked on both sides by ghosts. In my personal playthroughs of the game, this happened many times because I wasn't careful. Something that was not included in Jr. Pac-Man however was warp tunnels that would teleport the player to the side opposite of the maze in which the player entered it. This restricts the player's movement even more. I was not prepared for this, as I had not known the warp tunnel was omitted in this game.

However, there are bonus items in this game that were not present in its source material. Bonus items will appear that will bounce through the maze along with the player, something that bonus items never did in Pac-Man. In addition to bouncing through the maze, any Pellets that it rolls over are mutated, and cost more points when eaten. However,

eating them also slows down Jr. Pac. The bonus items also bounce toward a Power Pellet, which it detonates upon contact with it.

As before, obstacles and events influence what becomes the priority during play. Self-preservation, or high score? What if the player is working on getting to the next level with the highest score possible? A bonus item has spawned, and is mutating Pellets. This is good, since it allows the player a chance for a high score. However, if the player isn't careful it will destroy a Power Pellet, which the player needs in order to ensure safety against the ghosts, and move onto the next level. The mutated Pellets also make the player slower when eating them, which increases risk of being caught by the ghosts. Let's also say that the Power Pellet the bonus item will destroy is the last one. Will the player choose to collect the item now, or take their chances with a high score instead?

https://youtu.be/DI7w_1Z4a7g?si=hXMbp4SEqrm4qbob

Graphic And Audio Design

Visual design

Response prompt

Explain how a visual design choice affects the experience or communicates information during play. Use a specific example.

Submitted response

Given its source material, the visual design of the game is simple yet effective. The top-down view of the game allows for the player to easily see what is going on around them, past the walls that bind the player to set paths. The maze's color is also noticeably duller than Jr. Pac and the ghosts. It offers higher contrast between the player, the enemies, and the surroundings, helping the player to more easily see where they are in relation to the ghosts.

The ghosts also communicate through the way they are animated. Jr. Pac is a solid yellow, while the ghosts flash repeatedly. Besides providing differentiation between themselves and the player, they also help convey the fact that they are ghosts. Ghosts are technically intangible, unable to be touched. The flashing makes them appear see-through, just like a ghost might be like. And just as the original Pac-Man, eating a Power Pellet turns them blue, though they retain their flashing animation. In addition to this, when they begin returning to normal they begin flashing white as well. The pace of the white flashing quickens when they're closer to turning back to normal.

The heavy contrast of the ghosts then places blame on the player. That is to say, there are multiple things indicating what is a ghost and where they are, and whether they can be eaten or not. As an example, the player may claim that the ghosts snuck up on him without him noticing during his playthrough. The ghosts have very clear signs as to what they look like and where they are given the top-down view. If they player fails, it's because they weren't paying closer attention, not because the game isn't fair.

Sound design

Response prompt

Explain how music, sound effects, or their absence affect the experience or communicate information during play. Use a specific example.

Submitted response

Music is sparse in Pac-Man in general, and Jr. Pac-Man follows suit. The game has a jingle when the game begins, and the only sounds heard after that are the signature "waka-waka" sounds of Jr. Pac and the constant humming drones of the wandering ghosts. There is also the more frantic synth sounds that play whenever Jr. Pac eats a Power Pellet and can eat the ghosts. There is also the hum-fizzle sound of when you get caught by a ghost and lose a life.

There are also new sounds with the added Bonus Items. There is a brief sound queue when a Bonus Item is spawned in, and a constant subtle "thud" noise whenever it collides with a wall. Jr. Pac's "waka-waka" sound is also noticeably heavier when eating the mutated Pellets, adding another signifier to them besides their visual (albeit minimal) change and how he moves slower when eating them.

The new sounds with the Bonus Items may be especially exciting to high score chasers. The sound of the Bonus Item spawning in alerts them to the possibility of a higher score, either by collecting the item itself or letting it wander through the maze for a bit in order to mutate as many Pellets as possible. They may also be searching for a more exciting challenge: a slower movement speed and less Power Pellets to rely on.

Narrative And Storytelling

Narrative and setting

Response prompt

Explain how the setting and story relate to play. Describe key events if present, or explain the effect of a limited or absent story.

Submitted response

At its core, Jr. Pac-Man does not strictly have a story to follow. The setting of the main gameplay is in a maze, one with no warp tunnels that teleports Jr. Pac to the opposite side of the maze when entered. The maze changes color when entering a new level, some aspects of the maze changing shape as well.

There is, however, a short story told between levels involving Jr. Pac-Man and a ghost named Yum-Yum. The story features the two in a sort of Romeo and Juliet-type of story. While entertaining, it does not affect the gameplay all that much. Though it doesn't really need to, since the gameplay itself is straightforward. The setting is already simple: Jr. Pac-Man is the son of Pac-Man and Ms. Pac-Man, in a maze with ghosts that he needs to avoid in order to collect all the Pellets. Rinse and repeat until you reach the final maze, of which there are seven of.

Character development

Response prompt

Explain how character development relates to gameplay, or how limited or absent development affects the experience.

Submitted response

There isn't really any character development in the game itself. The only interactions there are is that the ghosts can kill Jr. Pac, and if Jr. Pac eats a Power Pellet then he can eat them instead, only for them to return shortly after to resume hunting him through the maze. The absence of any character development does not detract from the game, though. The goal of the game is so simple that it doesn't require any of the characters involved to improve or deteriorate.

However, there is slight character work in the cutscenes previously mentioned. The first cutscene that players experience is Jr. Pac meeting Yum-Yum for the first time. However, Ms. Pac-man and Yum-Yum's parent (the red ghost, Blinky) disapprove of their relationship. However, the next scene the player can find is Jr. Pac gifting Yum-Yum a red balloon. The final cutscene features them escaping, and beginning a romantic relationship together. While cute, it again doesn't serve any purpose to the game except for being entertaining and possibly making the player smile a bit more. Although the idea of the two escaping could be the implication that Jr. Pac escaped the mazes the player was going through, and in doing so finally united him and Yum-Yum, no longer bound by the game's rules.

Impact And Reception

Reception

Response prompt

Summarize the critical and player reception using identifiable reviews or feedback. Distinguish release-era reception from later reactions.

Submitted response

Upon initial release, Jr. Pac-Man received mixed reception, with a majority leaning towards poor, both critically and commercially. A very little amount of official home ports were available during the time, and the game itself was facing disconnect with the developer due to not being Namco property. Some stated that the game felt harder than the

previous Pac-Man games, being faster than other maze-chase titles. Some also argued that the Bonus Items were also creating clutter in a game that's meant to be simple.

Later reviews for the game were less harsh. As time went on, higher-energy gameplay (i.e. quicker) was a welcome addition to new games. The music used and the horizontal scrolling feature also merited the game a kinder reception in retrospective reviewers. Home conversions and other ports were also being enjoyed.

<https://indiegamerchick.com/2023/08/10/jr-pac-man-arcade-review/>

Industry and cultural impact

Response prompt

Explain the game's industry impact and cultural significance using research. Avoid claiming influence that the sources do not establish.

Submitted response

In terms of industry, Jr. Pac-Man is the first Pac-Man game to incorporate horizontal scrolling in its gameplay, allowing the mazes for the game to be 3 times wider than older Pac-Man titles. Also unique to its gameplay was the previously mentioned wandering Bonus Items rather than having static ones. Besides gameplay, the game was very impactful due to the fact that it was not created by Pac-Man's original creator, Namco, starting ownership and licensing disputes over the game. These disputes were messy overall, and strict self-publishing rules imposed by Namco were not lifted until 1990. Regardless of that however, units for Jr. Pac-Man were still released, with the majority of them being conversion kits to be used on older game cabinets. These were done rather than making entirely new machines.

Culturally speaking, the game was considered much more difficult than previous Pac-Man titles. While it initially received mixed reviews (of which most were negative), the harder gameplay has received praise later on by those who enjoy more of a gaming challenge. The game would eventually become available as a home port for only a select few home consoles, like the Atari 2600.

<https://www.rotheblog.com/2007/08/arcade/history-jr-pac-man-sideart-mappy-cabinet-registry/>

<https://indiegamerchick.com/2023/08/10/jr-pac-man-arcade-review/>

Reflection

Fun and experience

Response prompt

Explain what made the game enjoyable or frustrating for you, using a gameplay example.

Submitted response

The game itself is fun, which was something I expected. I initially picked the game because I believed it to be an easy one to research (which it, as explained in the next section, was not). However, I found myself enjoying it the more I played it. The game was more challenging than the original Pac-Man title, what with the faster enemy AI, the moving Bonus Items, the lack of warp tunnels, etc. And I enjoy the cutscenes that were included in the game about Jr. Pac and Yum-Yum. A personal take, I enjoy a fun and silly love story (even if it is a short one).

Learning and research

Response prompt

Reflect on what this case study taught you and how easy or difficult it was to find dependable information.

Submitted response

Searching for information about a game this old is very difficult. I did my best to avoid Wikipedia as a source for information, even going so far as to use the Fandom website as a source of information. Unsurprisingly, that was not very helpful. YouTube videos recapping the history of the game turned out to be the most helpful when gathering information about it. And thankfully, I was able to find websites that were a hearty source of information for the game.

Thankfully, the game received a decent amount of documentation, most likely due to the legal trouble it faced, and also thanks to the collectors of old games and their related media.